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ORIGINAL ARTICLE

Effects of transcranial direct current stimulation combined with gait-oriented motor training on disability, quality of life, and motor function in individuals with subacute stroke: a randomized controlled trial

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ABSTRACT

BACKGROUND: Stroke is a leading cause of long-term disability, with the sub-acute phase (1–6 months post-stroke) being critical for recovery. While gait-oriented motor training is widely used in rehabilitation, its efficacy may be enhanced by adjunctive neuromodulation through transcranial direct current stimulation (tDCS). However, evidence remains limited regarding the combined impact of these approaches on both functional and quality-of-life outcomes in sub-acute stroke patients.
AIM: To assess the effectiveness of tDCS combined with gait-oriented motor training on disability, quality of life, motor function, and balance in individuals with sub-acute stroke.
DESIGN: A prospective, parallel-group, randomized controlled trial.
SETTING: Outpatient rehabilitation clinics affiliated with King Khalid University, Saudi Arabia.
POPULATION: Fifty-seven participants (aged 18-80 years) with medically stable, first-ever ischemic or hemorrhagic sub-acute stroke and lower limb motor deficits.
METHODS: Participants were randomized into an intervention group (tDCS + gait-oriented motor training, N.=29) or a control group (gait-oriented motor training only, N.=28). tDCS was administered at 2 mA for 20 minutes, three times per week for four weeks. Primary outcomes were stroke-specific health-related quality of life (Stroke Impact Scale, SIS) and general functioning/disability (WHODAS 2.0), in line with WHO's ICF framework. Secondary outcomes included motor function (Fugl-Meyer Assessment), balance (Berg Balance Scale), and mobility (Timed Up and Go test). Assessments occurred at baseline, post-intervention, and at three-month follow-up.
RESULTS: The intervention group showed significantly greater improvements than the control group across all primary and secondary outcomes ($P<0.01$). Effect sizes were moderate to large (Cohen's d=0.44-1.2). Improvements in motor function and balance were positively correlated with gains in independence and social participation.
CONCLUSIONS: The combination of tDCS and gait-oriented training significantly improves motor function, participation, and self-perceived social and emotional functioning in sub-acute stroke rehabilitation and demonstrates sustained effects over time.
CLINICAL REHABILITATION IMPACT: This integrated intervention provides a viable and effective strategy for enhancing motor recovery, balance, and quality of life in stroke patients, supporting its incorporation into routine outpatient rehabilitation protocols.

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KEY WORDS: Stroke rehabilitation; Transcranial direct current stimulation; Recovery of function; Quality of life.

Stroke continues to be a leading cause of disability and death globally and represents a growing challenge for individuals and societies.[1] The first critical period for rehabilitation among stroke survivors is the subacute phase, which generally lasts from one to six months post-stroke.[2] This is the period during which neurophysiological recovery mechanisms, including spontaneous repair processes and neuroplasticity, are most active.[3] However, many people in the sub-acute phase are facing persistent motor, balance, and cognitive impairments and are disabled regarding their autonomy and quality of life.[4] Typical rehabilitation practices during this phase focus on physical function and mobility; however, achieving and sustaining improvements in psychosocial outcomes remain challenging.[5]

One such neuromodulatory intervention with tremendous potential for augmentation of post-stroke recovery is transcranial direct current stimulation (tDCS).[5] tDCS enhances neuronal excitability and induces neuroplasticity by passing a weak electrical current through the scalp to specific cortical regions.[6] It is especially effective when used alongside task-specific motor training since it primes neural circuits to enhance rehabilitation training.[6] Numerous studies have demonstrated that tDCS can facilitate motor relearning, decrease spasticity, and improve functional mobility in stroke patients.[7] While these findings are promising, the use of tDCS in the clinic is hampered by widely varying treatment protocols and outcomes.[7] The objective of this study was to fill this gap in knowledge by conducting a systematic evaluation of the effects of tDCS combined with gait-oriented motor training on disability and quality of life among patients with sub-acute stroke.[7] Seeking to demonstrate this with validated outcome measures, including the Stroke Impact Scale (SIS) and the World Health Organization Disability Assessment Schedule (WHODAS 2.0), this study aims to provide the first compelling evidence around the intervention's impact in these areas of marked importance to this population.[8]

Impairment of motor function and imbalance are among the most disabling sequelae of stroke.[9] These conditions are prevalent and significantly contribute to falls, a general decline in independence, and a diminished quality of life.[10] In order to facilitate functional recovery and reintegration into the community, interventions must address these deficiencies. A critical component of stroke rehabilitation is the task-specific and repetitive exercise of gait-related duties, which includes gait-oriented motor training.[11] By integrating this module with tDCS, the effects may be enhanced through a synergistic effect that involves both motor cortical excitability and peripheral motor performance.[12] This study employs validated clinical markers, such as the Timed Up and Go (TUG) test, the Berg Balance Scale (BBS), and the Fugl-Meyer Assessment (FMA) for lower extremities, to comprehensively evaluate these impacts.[13] These measures provide a thorough examination of the intervention's effects, establishing a solid foundation for assessing changes in functional mobility, postural stability, and motor control.[13]

However, there is still an unmet need to optimize interventions that target functional and quality-of-life outcomes in the sub-acute phase despite the advancements in stroke rehabilitation.[14] The overall lack of promising evidence on tDCS may be attributed to the non-comparability of studies, which is a result of heterogeneous electrode montage, stimulation settings, and outcome measures.[15] In addition, stroke-related disability is multidimensional, yet the majority of studies have only evaluated motor or cognitive recovery independently.[16] Comprehensive interventions that incorporate neuromodulation and task-specific training have to be implemented immediately to enhance both physical and psychosocial outcomes.[17] The present study addresses this lacuna by integrating tDCS into gait-oriented motor training and evaluating its impact on a diverse array of outcomes, such as balance, motor function, quality of life, and disability. It also provides a perspective on the long- and short-term effects of this combination intervention, as it addresses the knowledge deficit regarding the sustainability of improvements, which is exacerbated by the scarcity of longitudinal data.

The main aim of this study is to find out the effects of tDCS in combination with gait-oriented motor training on disability and quality of life of sub-acute stroke patients, assessed by specific questionnaires SIS and WHODAS 2.0 from the health-related quality of life assessments. The SIS was selected as a health-related quality of life instrument capturing stroke-specific domains, including mobility, participation, emotion, and cognition. In contrast, WHODAS 2.0 was chosen to evaluate global functioning and disability according to the ICF model, reflecting limitations in daily activities and social participation irrespective of diagnosis. The secondary objective is to evaluate the reliability of motor function and balance markers through clinical tests such as TUG, BBS, and FMA. The intervention group is expected to show a significantly greater improvement in disability, quality of life, motor function, and balance compared to the control group. Furthermore, these positive changes may maintain over the follow-up period, highlighting the recovery potential of this intervention in the long term.

Materials and methods

Design, settings, and ethics

This randomized controlled trial was performed from August 2023 to May 2024 in the Rehabilitation Medicine Department clinics, King Khalid University affiliated hospitals specializing in stroke rehabilitation, both situated in Saudi Arabia. This study was conducted in accordance with the principles of the Declaration of Helsinki for medical research involving human beings, and written informed consent was obtained from each participant before enrollment. The study received ethical approval from the Institutional Review Board of the Deanship of Scientific Research, King Khalid University (ECM#2019-31), on August 18, 2023. The trial registration number is ISRCTN62197971.

Participants

This study includes participants who were diagnosed with a sub-acute stroke, which is defined as the one to six-month period after the onset of stroke, based on their clinical evaluation and neuroimaging confirmation. Participants were recruited from the outpatient services of the Rehabilitation Medicine Department clinics of King Khalid University and its associated hospitals through a combination of referral screening and direct advertisement in the hospital. The inclusion criteria included medically stable adults aged 18–80 years who had experienced a first-ever ischemic or hemorrhagic stroke and had reduced motor control in the lower extremities, according to FMA.[18] Participants also underwent MMSE (Mini-Mental State Examination) to evaluate their cognitive function as inclusion criteria, and their score was ≥24.[18] Exclusion criteria included the presence of severe spasticity (Modified Ashworth Scale score >3), currently participating in another rehabilitation program, significant comorbidities that might cause exclusion from participation (severe cardiovascular disease or orthopedic conditions), and tDCS contraindications such as implanted electronic devices or history of epilepsy.[18] Also excluded were individuals with severe aphasia or cognitive deficits that would prevent them from providing informed consent or compliance with tasks. The sample was selected in a systematic way to ensure a representative cross-section of the sub-acute stroke population. A clinical assessment confirmed participant eligibility. Written informed consent was obtained after explaining the study objectives, protocols, and potential risks.

Intervention

A curated protocol with tDCS and gait-oriented motor training was applied to the intervention group, where tDCS was the neuromodulatory component applied to increase cortical excitability.[19] tDCS was administrated by Soterix Medical 1x1 tDCS device, a commonly applied device that was able to return a steady output current of 2 mA (Figure 1).[20] All sessions lasted 20 min, as this duration has been demonstrated to maximize the neuromodulatory effects while minimizing unwanted side effects. Parameters of the stimulation (current intensity and duration) were selected according to the existing literature for the maximal effects on neuroplasticity and functional recovery in post-stroke patients.[21] The electrodes were placed following the International 10-20 EEG system in order to properly target the different brain regions.[21] The anode electrode (5×7 cm) was placed on the primary motor cortex (M1) of the affect-

Figure 1.—Implementation of transcranial direct current stimulation within a standardized neuromodulatory protocol for the intervention cohort.

ed hemisphere, located at C3 or C4, depending on the side of the lesion.[21] This montage – anode over ipsilesional M1 and cathode over the contralateral supraorbital area – was selected based on its consistent use in post-stroke neuromodulation protocols targeting lower limb recovery. It is supported by meta-analyses and controlled trials showing that this configuration promotes corticospinal excitability in the affected hemisphere while minimizing inhibitory interference from the non-lesioned side. This electrode setup is also better tolerated and has been linked with improved gait and mobility outcomes in subacute stroke cohorts. This approach was meant to promote excitability of the motor areas of the contralateral lower extremities. To complete the circuit and avoid interference with other cortical regions, the cathode electrode (5×7 cm) was placed over the contralateral supraorbital region, which acted as the reference electrode.[21] Saline-soaked sponges housed the electrodes to provide optimal current delivery and participant comfort. To improve the conductivity of the sponge material, a standard saline solution was saturated in the sponge material to reduce the chance of skin irritation.[22] Each session lasted for 20 minutes s with a 30-second ramp-up phase [where the current intensity increased from 0 to 2 mA over 30 s] in the beginning to minimize any transitory discomfort usually associated with the onset of stimulation.[21] Likewise, a ramp-down phase was performed at the end of each session to gradually decrease the current, which improved tolerance to the procedure.[21] The tDCS sessions were delivered three days a week, four weeks in total, in line with established clinical protocols shown to be effective in stroke rehabilitation. A trained rehabilitation specialist supervised each session to ensure the correct position of the electrodes, optimal functioning of the device, and safety of participants. The protocol was closely monitored, and participants were asked to collect and report any adverse events, like mild skin irritation or transient headache. However, no adverse events were reported throughout the study period. All tDCS sessions were standardized to allow for consistency between sessions, contributing to the reliability of the intervention and its outcomes.

After the tDCS intervention, the intervention group performed a gait task-oriented motor learning protocol targeting the key components of functional mobility, balance, and lower extremity strength. The training protocol followed evidence-based principles of task-specific and repetitive practice to help promote motor relearning and neuroplasticity in stroke rehabilitation.[23] A series of 60-minute sessions that contained a multitude of exercises to suit the needs and capabilities of every individual. Treadmill gait (body-weight support) was one of the basic components in an environment under controlled conditions to promote repetitive gait cycles.[24] This device used a harness system to offload some of the participant's body weight, permitting safer walking mechanics practice and allowing individuals with limited mobility to achieve a normal gait pattern.[24] The study also provided overground gait training to reinforce the improvements achieved on the treadmill and to enhance the ability to walk independently in the community.[25] Participants practiced walking in a straight line on flat surfaces, walking between obstacles, and walking at different speeds to improve adaptability. A fundamental part of the regimen was balance exercises, which focused on decreasing fall risk and enhancing postural control. Static exercises included standing balance on stable and unstable surfaces, and dynamic tasks included weight-shifting, reaching, and step exercises.[26] By requiring participants to balance during these tasks, greater balance in movement was encouraged. Participants performed functional mobility tasks, such as sit-to-stand and stair climbing, targeting common limitations in daily activities among stroke survivors. Sit-to-stand specimens focused on lower extremity strength and movement efficiency, while stair climb-

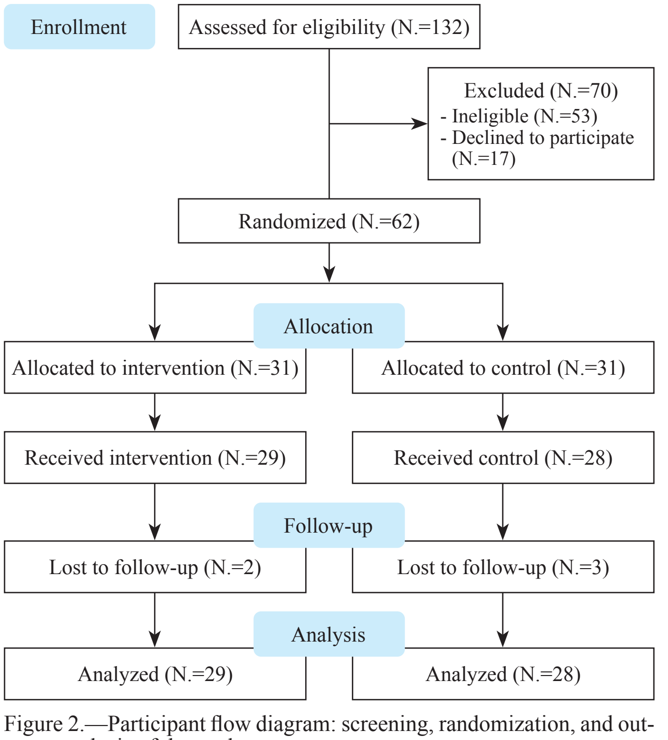

Figure 2.—Participant flow diagram: screening, randomization, and outcome analysis of the study.

ing was designed to develop the strength and coordination for more complex functional activities. These tasks were gradually compounded through increased demands (by increasing the number of steps to be climbed or the level of assistance to be reduced) so that the difficulty was adjusted as a function of the participant's performance in order to promote continual progress.[26] The training program included an individualization component. Before the intervention began, each participant's baseline capacity to function was assessed in order to generate an individualized training plan. Session progress was tracked, and exercises were adjusted to ensure appropriate levels of challenge whilst maintaining safety, progression, or maintenance orientation. Participants were closely monitored and supervised by trained rehabilitation therapists to maintain the correct technique, avoid injuries, and enhance the training efficacy.

The control group received only gait-oriented motor training, matched for duration, frequency, and intensity as was provided for the intervention group. To maximize protocol compliance, both groups were supervised by trained rehabilitation therapists. In both groups, sessions were held three days a week for four weeks. Participants were monitored for adverse events. Outcome assessments were conducted by assessors blinded to group allocation at baseline, immediately after the intervention, and at the three-month follow-up.

Outcomes

Primary study outcome measures were specifically defined to assess intervention effects in functional and quality of life domains among sub-acute stroke patients.[27] The main outcomes were stroke-specific health status and global functioning/disability, measured with the Stroke Impact Scale (SIS)[28] and the World Health Organization Disability Assessment Schedule (WHODAS 2.0),[29] respectively. SIS assesses stroke-related health-related quality of life across eight domains, including physical function, emotion, and social participation. The SIS consists of 59 items across 8 domains: strength, hand function, ADL/IADL, mobility, communication, emotion, memory and thinking, and participation. Each item is rated on a 5-point Likert scale. Domain scores are transformed into a scale from 0 to 100, where higher scores indicate better perceived function. The SIS total score is the average of domain scores and reflects overall stroke-specific health status, with higher values indicating lower disability and better quality of life. WHODAS 2.0 captures functional limitations and disability in daily activities and participation based on the ICF conceptual model.[28] WHODAS 2.0 includes 36 items across 6 domains: cognition, mobility, self-care, getting along, life activities, and participation. Each item is scored on a 5-point scale, and the total score is calculated using item-response theory or simple summation. Scores range from 0 to 100, with higher scores indicating greater disability and lower functioning across daily and social activities.[29] The assessments were conducted in a quiet clinical setting by trained evaluators to minimize deviations in administration and response, and both scales have been validated for use in stroke rehabilitation literature. The secondary outcomes focused on motor function, balance, and functional mobility, measured with the Fugl-Meyer Assessment for Lower Extremities (FMA-LE),[30] the BBS[31] and the TUG test.[32] The FMA-LE is a gold standard assessment tool used to measure motor recovery in the affected lower limb, examining strength, coordination, and reflexes in stroke rehabilitation literature.[30] The BBS evaluated subjects' capacity to hold their balance while performing functional tasks, such as standing, reaching, and transferring, supplying a detailed measure of postural control and risk of falls.[31] The TUG test was performed to evaluate dynamic balance and functional mobility, which needed the participants to stand from a chair, walk a predetermined distance, turn, and return, representing a practical assessment of walking ability and mobility.[32] All outcomes were assessed at three time points (baseline, post-intervention [Week 4], and follow-up [Month 3]). These outcome measures were selected to capture recovery across all domains of the ICF model—participation (SIS, WHODAS 2.0), activity (BBS, TUG), and impairment (FMA). SIS and WHODAS 2.0 were prioritized as primary outcomes due to their validated use in post-stroke populations and their emphasis on real-life function, participation, and quality of life. While these tools are commonly used in clinical practice, they have also demonstrated sensitivity to change and relevance in stroke rehabilitation research. Secondary measures were chosen to evaluate physical function more directly and triangulate recovery trends through both clinical and functional lenses.

Sample size estimation

The study determined the size of the sample for this study using the G*Power 3.1 software to ensure that we would be able to demonstrate a statistically significant clinically meaningful difference between the intervention and control groups. This estimate was based on a previously reported expected effect size (Cohen's dd) of 0.7 for tDCS efficacy in stroke rehabilitation.[33] Sample size was determined by using 0.05 for significance level (α) and 0.80 for statistical power (1-β1) analysis using a two-tailed independent samples *t*-test. This design controlled for variability of the primary outcome, disability, and quality of life, as measured by the SIS. Based on these parameters, the estimated sample size required was determined to be 26 participants per group, totaling 52 participants. To address possible dropouts, an over-enrollment of 10% was added (N.=62). The calculation was based on detecting a between-group difference in total SIS score, with an anticipated effect size (Cohen's *d*) of 0.7 derived from previous studies evaluating tDCS effects on functional recovery in stroke. A minimum clinically important difference (MCID) of approximately 10 points in the SIS total score was used as a reference for expected change. The analysis was conducted using G*Power 3.1, applying a two-tailed independent samples *t*-test, with alpha=0.05 and power=0.80.

Allocation, randomization and implementaiaton

To minimize selection bias, randomization was performed to allocate the total sample of 62 participants equally between the intervention and control groups. To ensure equal distribution of the groups in the study, a computer-generated block size six random allocation sequence was prepared by permuted block randomization approach. Each block had three people assigned to be in the intervention group and three in the control group. The randomization sequence was generated by an external statistician not involved in recruiting, delivering the intervention, or assessing outcomes to allow for bias-free randomization and the methodological soundness of the trial. Allocation concealment was accomplished with sealed, opaque, and consecutively numbered envelopes, each containing the group assignment according to the randomization sequence. The envelopes were held by an independent administrator unaffiliated with the research team. An envelope was opened in the participant's presence by the administrator upon confirmation of eligibility and receipt of informed consent. This method kept informed center groupings hidden until the time of allocation and prevented any bias in a participant assignment. The independent administrator conducted the randomization process according to the allocation sequence that was defined before the beginning of the study, completing all blocks without adaptation. To minimize potential performance and detection bias, investigators and outcome evaluators were masked to group assignment for the entire study.

Blinding

Blinding was adopted to reduce bias and improve the methodological quality of the study. Participants and care providers administering the interventions were not blinded, given that blinding of treatment assignments was impractical due to the nature of the interventions. Outcome assessors (blinded to group assignment) assessed the primary and secondary outcomes, thereby preventing detection bias. All assessments were performed outside of the intervention sessions to minimize blinding, and participants were directed not to communicate their group allocation to the assessors. The control group was given gait-oriented motor training in terms of structure, duration, and intensity, similar to that of the intervention group, except for the application of tDCS. Such a non-tDCS component in both intervention groups ensured uniformity in terms of anything that was not due to tDCS (training), allowing us to attribute differences in outcomes directly to tDCS components of the intervention.

Statistical analysis

Data were processed using SPSS version 24, and a P value of <0.05 was regarded as statistically significant. For the primary aim, a Multivariate Analysis of Covariance (MANCOVA) was performed to compare disability and quality-of-life scores between the intervention and control groups, adjusted for baseline differences in these variables. Also, longitudinal mixed-effects models were used to assess changes in disability and quality of life across three time points (*i.e.*, baseline, post-intervention, and 3-month follow-up) to examine both immediate and delayed effects. For the secondary endpoint, repeated measures ANOVA examined differences in motor function and balance scores over time points and between groups. The study conducted a Pearson correlation analysis to explore the relationships between the improvements in motor function and the changes in quality of life. The Shapiro-Wilk Test indicated that the data adhere to a normal distribution, which permits the application of parametric methods for statistical analysis.

Results

Out of 132 participants screened, 62 were randomly assigned to the intervention and control groups in equal measure. After accounting for minor losses during follow-up, a total of 29 participants in the intervention group and 28 in the control group were analyzed, demonstrating strong recruitment and retention as illustrated in the participant flow diagram (Figure 2).

Recruitment for the study occurred from August 2023 to May 2024, with follow-up assessments finalized by March 2024. The trial was executed as intended and was not terminated early, as the target sample size of 62 participants was reached within the specified recruitment timeframe. Participants were monitored for three months following the intervention to evaluate the persistence of the observed results. The baseline characteristics of individuals in the intervention and control groups were predominantly similar, with no significant differences noted in factors like age, gender distribution, BMI, hypertension, diabetes, smoking status, dyslipidemia, and prior stroke history (Table I). Notable disparities were observed in baseline quality of life scores (WHODAS, P=0.005), motor function scores (FMA, P=0.015), and balance scores (BBS, P=0.002), with the intervention group exhibiting marginally elevated initial values. The time elapsed since the stroke varied considerably (P=0.042), indicating differences in recovery timelines among groups.

Improvements in disability and quality-of-life outcomes for the intervention group were significantly greater than those for the control group and presented as significantly higher SIS total scores (101.7 87.5) and quality-of-life (WHODAS) scores (18.2 *vs.* 57.7) both postintervention and at follow-up (Table II). Similarly, effect sizes for disability (Cohen's d=1.2) and physical function (Cohen's d=0.7) indicate clear clinical benefit. There were significant improvements in the mobility (P=0.001) and participation (P=0.001) domains of the SIS, which also indicates the success of the intervention.

The intervention group showed clinically important improvement in motor function, balance, and functional ability compared with the control group (P<0.01), with the differences significant (mean difference 9.75 CI (95%), 9.67–9.84), to post-intervention, and follow-up (Table III) without CTRS and TUG. The intervention group demon-

Table I.—*Baseline demographic and clinical characteristics of participants in intervention and control groups.*

Variable	Intervention group (mean±SD or %)	Control group (Mean±SD or %)	P value
Age (years)	61.30±7.50	60.10±7.85	0.215
Gender (Male/female)	15/14 (51.7% Male)	14/14 (50.0% Male)	0.850
BMI (kg/m^2)	25.1±2.4	24.9±2.5	0.760
Duration since stroke (months)	6.10±1.45	5.55±1.60	0.042
Affected side (left/right)	13/16 (44.8% Left)	12/16 (42.9% Left)	1.000
Hypertension (yes/no)	18/11 (62.1%)	17/11 (60.7%)	1.000
Diabetes (yes/no)	10/19 (34.5%)	9/19 (32.1%)	0.840
Smoking (yes/no)	8/21 (27.6%)	7/21 (25.0%)	1.000
Dyslipidemia (yes/no)	9/20 (31.0%)	8/20 (28.6%)	0.835
Previous stroke (yes/no)	2/27 (6.9%)	3/25 (10.7%)	1.000
Baseline Disability Score (SIS)	50.80±9.8	47.20±9.9	0.170
Baseline Quality of Life Score (WHODAS)	39.5±8.2	38.10±7.8	0.005
Baseline Motor Function Score (FMA)	20.30±5.2	18.60±6.0	0.015
Baseline Balance Score (BBS)	25.10±4.6	25.80±4.7	0.002
Baseline Spasticity Score (Modified Ashworth Scale)	1.95±0.40	2.10±0.42	0.490
Baseline Depression Score (PHQ-9)	9.75±2.4	9.30±2.8	0.002

BMI: Body Mass Index; SIS: Stroke Impact Scale; WHODAS: World Health Organization Disability Assessment Schedule; FMA: Fugl-Meyer Assessment; BBS: Berg Balance Scale; PHQ-9: Patient Health Questionnaire-9; Modified Ashworth Scale: a scale for spasticity.
Baseline differences in WHODAS, FMA, and duration since stroke were adjusted using MANCOVA to control for confounding factors and ensure accurate comparisons of intervention outcomes.

Table II.—*Effects of intervention on disability and quality of life outcomes.*

Variable	Baseline (mean±SD, intervention group)	Post-intervention (mean±SD, intervention group)	Follow-up (mean±SD, intervention group)	Baseline (mean±SD, control group)	Post-intervention (mean±SD, control group)	Follow-Up (mean±SD, control group)	Effect size (Cohen's d)	P value
Disability Score (SIS Total)	49.75±9.88	69.04±9.54	70.6±10.64	47.53±9.03	58.1±8.94	58.82±8.88	1.2	0.001
Quality of Life Score (WHODAS Total)	40.42±8.3	50.95±7.07	52.62±6.1	37.74±8.07	46.51±9.11	47.29±8.27	0.44	0.003
Physical Function Domain (SIS)	44.45±6.47	72.77±9.51	73.93±7.69	42.43±7.03	64.75±8.86	66.64±8.98	0.7	0.013
Mobility Domain (SIS)	41.29±6.62	69.32±8.95	70.34±9.72	41.47±5.9	62.6±11.18	65.22±10.08	0.45	0.001
Participation Domain (SIS)	36.69±8.34	58.86±9.39	63.34±6.91	34.79±7.11	53.06±7.87	55.87±7.23	0.67	0.001

SIS: Stroke Impact Scale; WHODAS: World Health Organization Disability Assessment Schedule; CI: confidence interval; SD: standard deviation.

Table III.—*Effects of intervention on motor function, balance, and functional mobility.*

Variable	Baseline (mean±SD, intervention group)	Post-intervention (mean±SD, intervention Group)	Follow-Up (mean±SD, intervention group)	Baseline (mean±SD, control group)	Post-intervention (mean±SD, control group)	Follow-up (mean±SD, control group)	Effect size (Cohen's d)	P value
Motor Function (FMA Total Score)	25.43±4.63	38.94±4.36	41.13±4.98	25.11±4.49	34.0±4.95	35.9±4.67	1.0	0.001
Balance (BBS Total Score)	29.77±5.85	49.57±5.99	49.82±6.19	28.68±6.01	44.85±6.1	46.06±5.79	0.83	0.002
Functional Mobility (Timed Up and Go seconds)	15.06±1.28	11.95±1.27	11.74±0.95	15.58±1.62	13.8±1.72	13.94±1.45	1.11	0.003

FMA: Fugl-Meyer Assessment; BBS: Berg Balance Scale; Timed Up and Go, a measure of functional mobility; SD: standard deviation.

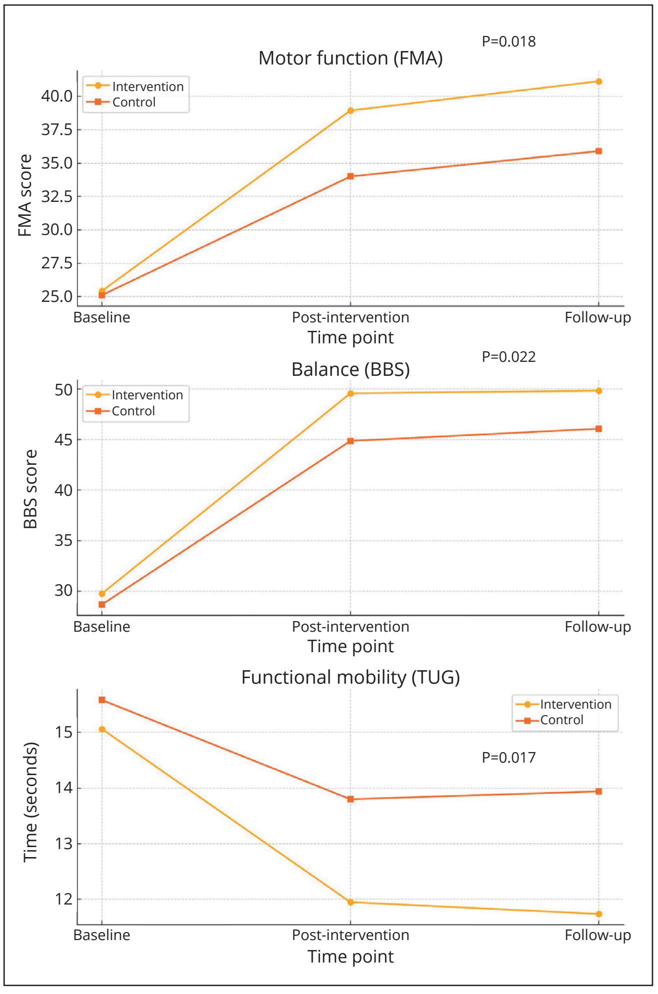

Figure 3.—Time × group interaction effects of tDCS combined with gait-oriented motor training on motor function, balance, and functional mobility in subacute stroke patients.

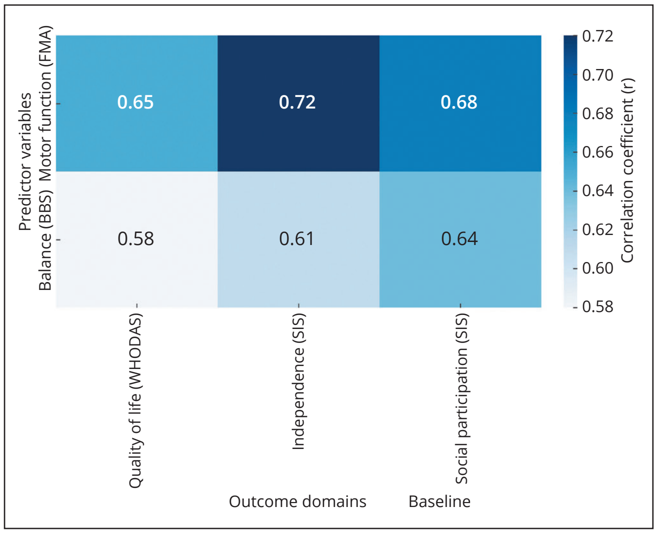

Figure 4.—Heatmap: correlation between motor function, balance, and quality of life domains in sub-acute stroke patients.

strated greater and more sustained changes over time than the control group, as evidenced by significant time × group interactions for motor function, balance, and functional mobility (Figure 3). This significantly improved motor function (FMA scores, P=0.018), balance (BBS scores, P=0.022), and reduction in functional mobility time (TUG scores, P=0.017), indicating the efficacy of the intervention in long-term improvement.

Results noted significant positive correlations between motor function, balance, and different quality of life domains, confirming that clinically relevant effects in terms of functional improvement were associated with better scores in patient-reported outcomes (Figure 4). Motor function (FMA) correlated most strongly with independence (r=0.72, P=0.001) and with social participation (r=0.68, P=0.003), whereas balance (BBS) was correlated with quality of life (WHODAS, r=0.58, P=0.005) and independence (r=0.61, P=0.004).

Discussion

This study aimed to examine the effect of combined tDCS with gait-oriented motor training on disability, quality of life, motor function, and balance in patients with sub-acute stroke. The results show that the intervention resulted in statistically significant improvements in all primary and secondary outcomes as compared to the control group. These results were associated with improvements in stroke-specific health-related quality of life (SIS) and global functioning and disability (WHODAS 2.0), aligning with both condition-specific and generic recovery indicators. The results indicate that the joint intervention improves functional rehabilitation and psychosocial function in patients with stroke.

This study demonstrates significant evidence of the synergistic effects of tDCS in conjunction with gait-oriented motor training in improving the multifactorial deficits observed in sub-acute stroke patients.[34, 35] This intervention produces synergistic effects that augment both neuroplasticity and functional recovery.[35] tDCS can increase or decrease excitability at the cortical level, enhancing synaptic plasticity to establish a permissive environment for motor relearning and promote cortical circuit reorganization.[36] Neuromotor entrainment combines sensory input with movement to enhance neural priming. Gait-oriented motor training incorporates task-specific, repetitive movements to strengthen motor pathways, improving physical function.[37] Collectively, these elements create a feedback loop that enhances recovery in motor function, balance, and mobility.[37] The continued improvements in both disability and quality-of-life outcomes highlight the intervention's capacity to facilitate long-term recovery, targeting both neurological impairments and patient-reported domains such as participation and emotional well-being.[38]

These results are consistent with earlier studies which have proven the effectiveness of such interventions in stroke rehabilitation. However, not all studies have shown benefit from tDCS in similar populations. For instance, Leon *et al.*[39] conducted a randomized controlled trial with a sham tDCS arm and reported no added benefit of tDCS for lower limb recovery. Differences in study design, including timing post-stroke, patient characteristics, stimulation parameters (montage, intensity, duration), and the type or intensity of motor training may explain these divergent outcomes. Our study focused on the subacute phase and employed a high-intensity, task-specific gait protocol, which may have better synergized with tDCS to facilitate neuroplasticity. Dey *et al.*[40] underscored the efficacy of task-based motor training paired with neuromodulatory techniques like tDCS, which drive substantial transfer of motor, functional, and postural gains, demonstrating that combining neuromodulation with physical training is beneficial.[41] Additionally, Navarro-Lopez *et al.*[42] demonstrated that the combination of tDCS and rehabilitative movement exercises was the most effective treatment for improving functional independence and mobility when compared to standard therapy, strengthening our findings regarding improved motor recovery and mobility.[42] Furthermore, Criekinge *et al.*[43] Thus, the findings from this study further highlight the clinical significance of the relationships between balance, motor function, and quality of life. Moreover, this evidence provides additional motivation on why combined neuromodulation and motor training protocols should be preferred in this context, just as it has been shown to yield clinically relevant effects on stroke rehabilitation.[44] However, other trials, such as Leon *et al.*,[39] reported no significant benefit from adding tDCS in a similar stroke population using a sham-controlled design. Differences in clinical phase (subacute *vs.* chronic), stimulation dose, electrode montage, and intensity of motor training may explain this discrepancy. Our study focused on the subacute phase, when cortical plasticity is heightened, and implemented a structured, individualized gait training protocol, potentially enhancing responsiveness to tDCS. The timing and context of intervention delivery likely influence interstudy variability.

The well-balanced functional assessments underpin the holistic view of recovery and its effect on patient-reported outcomes in this stroke-rehabilitative period, considering the strong positive correlations between motor function, balance, and quality-of-life domains. Functional movement scores, measured with the FMA, were the most significantly associated with social participation and independence, which highlights the importance of this as an outcome of rehabilitation, both in fulfilling the daily demands of life and the beneficial effect on returning to society.[45] These data demonstrate that improvement in motor control can, in the best of circumstances, lead to real and meaningful changes in the quality of life in patients.[46] Likewise, correlations between balance (as shown in BBS scoring) and quality-of-life domains such as independence and social participation denote that postural stability is indispensable to reaching copyright achievements.[47] Such associations suggest that any one motor function and balance intervention provokes trans-dimensional change, improving physical, emotional, and social aspects of well-being and facilitating more widespread recovery.[48] While the study did not strictly adopt a hierarchical ICF outcome structure, the design was conceptually aligned to cover impairment (FMA), activity (BBS, TUG), and participation (SIS, WHODAS). This was a deliberate choice to reflect real-world multidimensional recovery outcomes in sub-acute stroke, though we recognize the value of more rigid ICF mapping in future designs.

The results observed are similar to other studies. For instance, Anouk *et al.*[49] tend to be significantly associated with improved motor function and better quality of life, elaborating on the importance of motor recovery in patient-reported outcomes.[49] Similarly, Rafsten *et al.*[47] showed that improvements in balance and postural control were associated with better independence and social participation, which is also in line with the present study results. Furthermore, Kwakkel *et al.*[50] and the findings of Gomes *et al.*[51] emphasized that rehabilitation interventions that targeted both motor function and balance had better patient-reported outcomes and may be fundamental contributors to success in rehabilitation.[51] The significant associations found in our current work justify the need for further integration of motor and balance-focused therapies in stroke rehabilitation to maximize physical recovery and quality of life.[52]

Clinical significance

This study's findings provide sufficient evidence to include tDCS with gait-oriented motor training as a comprehensive rehabilitation strategy in one of the important populations, like sub-acute stroke individuals. The intervention resulted in significant improvements in mobility, balance, and motor function as well as significant improvements in participation, emotional functioning, and self-perceived disability, with enduring benefits. The solid correlations of motor and balance recoveries with important quality-of-life domains (independence and social participation) highlighted by this study underline the comprehensive impact of this innovative intervention on both physical and psychosocial rehabilitation endpoints. It offers to solve some of the multifaceted problems entangled with stroke rehabilitation and potentially become an economical treatment to be practiced in clinics. These may correlate to functional independence, community reintegration, and further patient outcomes.

Limitations of the study

Despite the encouraging results, there are multiple limitations to this study that need to be acknowledged. First, the limited follow-up period does not enable a full assessment of the sustainability of the observed improvements in motor function, balance, and quality of life over the long term. Although the sample size is adequately powered, the scope of responses does not fully capture the variability expected in a heterogeneous stroke population. Additionally, significant baseline differences between groups were observed in WHODAS, FMA, BBS scores, and time since stroke onset. Although these were adjusted using MANCOVA and mixed models, the possibility of residual confounding remains and may have influenced outcome interpretation. Future studies should consider stratified randomization or covariate-constrained allocation to better balance baseline characteristics across groups. Third, while clinical measures have been validated, they may not truly mirror neurophysiological changes within the host neurophysiology that can be explored with advanced imaging or electrophysiological approaches in subsequent investigations. Moreover, in the study, the parameters of tDCS were not varied, nor was the relative contribution of each of the components of the intervention explored, such as intensity and duration of motor training. Future research should aim to overcome these limitations by exploring the long-term effects of this intervention, optimizing treatment parameters, and investigating the application of this intervention to a wider and more heterogeneous stroke population. Another limitation is the absence of a sham tDCS control group. Although both groups received equal treatment duration and therapist contact time, the lack of sham stimulation precludes definitive conclusions about the extent to which observed effects were driven by neuromodulatory mechanisms *versus* placebo or expectation effects. Future research should incorporate a sham tDCS arm to control for these factors and enhance internal validity. There were also statistically significant baseline differences in WHODAS, FMA, BBS scores, and time since stroke between the two groups. While these differences were adjusted using MANCOVA and mixed-effects modeling, residual confounding cannot be completely excluded. This should be considered when interpreting the magnitude and direction of treatment effects, and future studies should consider stratified randomization or covariate-constrained allocation to improve baseline comparability. Furthermore, the magnitude of improvement observed in WHODAS scores – though statistically significant – may partially reflect expectancy effects, as the instrument is entirely self-reported. The lack of a sham condition limits our ability to isolate the neuromodulatory effects of tDCS from psychological or perceptual influences. Future studies should implement placebo-controlled conditions to mitigate reporting bias and validate the true clinical magnitude of change. Incorporating neurophysiology assessments and personalized interventions within a neurorehabilitation framework may better demonstrate the clinical utility of this approach.[53] The observed reduction in WHODAS scores, though statistically significant, appears large relative to the short intervention period and may reflect expectancy effects or reporting bias. Since WHODAS is a self-reported tool, the lack of blinding to tDCS exposure could have amplified participants' perceptions of improvement. These factors introduce uncertainty about the magnitude of true clinical change. Future trials should include sham-controlled arms and additional objective outcome measures to validate self-reported gains.

Conclusions

In conclusion, our findings indicate that the combination of adjunctive tDCS and gait-oriented motor training has the potential to markedly enhance motor function, balance, functional mobility, and overall quality of life in individuals recovering from sub-acute stroke. Findings: The intervention resulted in significant improvements in stroke-specific health status (SIS) and reductions in general disability (WHODAS 2.0), with lasting physical and psychosocial benefits. The observed strong correlations between functional recovery and quality-of-life outcomes may indicate the broader impact of this intervention on patients. The findings underscore the potential of this integrated approach as a viable and scalable rehabilitation method for facilitating thorough recovery in sub-acute stroke patients.

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Conflicts of interest
Ravi S. Reddy and Jaya S. Tedla have given substantial contributions to the conception and design of the manuscript; Ajaya K. Midde and Venkata N. Kakaraparthi contributed to acquisition, analysis, and interpretation of the data. All authors participated in drafting the manuscript; Ravi S. Reddy revised it critically. All authors contributed equally to the manuscript and read and approved the final version of the manuscript.

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